



A comparative study of different machine learning methods coupled with GIS for landslide susceptibility assessment: a case study of N'fis basin, Marrakesh High Atlas (Morocco)

Hassan Ait Naceur¹ · Brahim Igmoulan¹ · Mustapha Namous² · Mostafa Amrhar³ · Omar Bourouay¹ · Mustapha Ouayah² · Mourad Jadoud⁴

Received: 8 March 2022 / Accepted: 18 May 2022 / Published online: 1 June 2022
© Saudi Society for Geosciences 2022

Abstract

Improving the predictive accuracy of models based on machine learning techniques for assessing landslide susceptibility is an area that attracts researchers' attention. In this study, we applied, evaluated, and compared three machine learning models: random forest (RF), artificial neuron network (ANN), and decision tree (DT), for spatial prediction of landslide susceptibility in the N'fis watershed located in the High Atlas (Morocco). A database from an inventory of 156 historical landslides, randomly split into a training (70%) and validation (30%) set, was combined with thematic maps of fourteen causal factors (elevation, slope angle, slope aspect, curvature, lithology, land use, distance to faults, distance to rivers, distance to roads, topographic position index, topographic wetness index, precipitation, soil type, and normalized vegetation index) to build and validate the models. The performance of these models was evaluated using several statistical indices and the ROC curve method. The results show that all three models have worked well. The analysis of these results indicates that the RF model is the best performing (AUC = 96.75%), followed by ANN (AUC = 94.40%), and finally DT (AUC = 85.65%).

Keywords Machine learning · Landslide susceptibility assessment · Factor importance · GIS · N'fis watershed

Introduction

A landslide is a natural catastrophe that happens when gravity forces a mass of debris, soil, or rock to fall down a slope (Park and Kim 2019). Landslides are natural disasters that can cause very serious economic losses and many deaths around the world. Froude and Petley (2018) reported global

occurrence records of 4862 landslides that caused 55,997 deaths during the period 2004 to 2016. Between 2014 and 2018 these hazards caused 4914 deaths, 27,110 people homeless, and 2.1 billion dollars in economic losses (Pham et al. 2020). Landslides are most common in Morocco's Rif Mountains, which are located in the country's northwestern area (Elmoulat et al. 2015). This risk also affects the regions of the High Atlas where about twenty people died as a result of a landslide in the Ijoukak region on July 24, 2019. Therefore, mapping areas vulnerable to landslides is an essential means of preventing and reducing potential losses.

To mitigate the negative impact of landslides on a region, it is critical to determine susceptibility to this hazard using a causative factor analysis based on scientific knowledge and to pre-manage sensitive regions where landslides may occur (Kimet al. 2018). Various researchers define landslide susceptibility as the spatial distribution of the probability of landslides occurring (Constantin et al. 2011). Several susceptibility models may be built based on the given data to estimate the probability of potential landslides. Various methods can be used to generate landslide susceptibility: expert criteria-based direct mapping (Salvatici et al. 2018).

Responsible Editor: Biswajeet Pradhan

✉ Hassan Ait Naceur
hsngeo@gmail.com

- ¹ Laboratory of Georesources, Geoenvironment and Civil Engineering (L3G), Faculty of Sciences and Techniques, Cadi Ayyad University, Marrakesh, Morocco
- ² Data Science for Sustainable Earth Laboratory (Data4Earth), Sultan Moulay Slimane University, Beni Mellal, Morocco
- ³ Faculté Des Sciences, 2390, 40000 Semlalia Marrakech, BP, Morocco
- ⁴ Laboratory Geosciences and Environmental Techniques, Faculty of Sciences El Jadida, Chouaib Doukkali University, El Jadida, Morocco

These maps are based on the analyst's assessments and are subjective. Stability evaluations based on mathematical correlations between resisting and driving forces that take into account a variety of elements such as slope geometry and geotechnical features of rocks and soils are known as deterministic models (Yilmaz 2009). These models can be used efficiently on a very small scale. However, one of the disadvantages of these methodologies is the high cost and difficulty of obtaining hydrological and geotechnical data, especially when studying broad and varied areas. The heuristic technique involves estimating the relative contributions of a variety of parameters to landslide generation and developing susceptibility classifications. The main limitation of these methods is the susceptibility assessments subjectivity.

Quantitative methods for landslide susceptibility assessment are divided into three categories: physics-based models, statistics-based correlation analysis, and machine learning approaches. On a regional level, physical-based models require thorough geotechnical, engineering, and geological data on site-specific slope failure. This type of method is unsuitable for assessing landslide susceptibility in a large spatial area, given the large number of soil parameters required.

Statistical techniques, along with remote sensing (RS) and geographic information systems (GIS), have been widely employed in recent decades. For instance, consider the frequency ratio (FR) (Rasyid et al. 2016; Chen et al. 2016a, b), information value (SI) (Manchar et al. 2018), as well as the weight of evidence (WoE) (Rana et al. 2017; Guo et al. 2015). Statistical models are based on the use of inventories of past landslides and other causative factors (such as slope aspect and degree, precipitation, etc.) that can be extracted on a large scale from Geographic Information Systems (GIS) and remote sensing data. These techniques have made the assessment of landslide susceptibility as an easy task for mapping landslide-prone areas on a large scale, where there is not enough geotechnical information available to make physics-based models (Al-Najjar and Pradhan 2021). The success of these methods is also due to the availability and accessibility of information offered by remote sensing, such as topography, terrain morphometry, and precipitation, which improves the application of large-scale statistical models.

Recently, the prediction of landslide susceptibility has been developed by introducing other more efficient methods of machine learning (ML) to determine with high precision the landslide-prone areas. In landslide susceptibility assessment, support vector machine-SVM and other machine learning models were largely used (Pourghasemi and Rahmati 2018; Kumar et al. 2017) naïve Bayes-NB (Chen et al. 2018a, b), boosted regression tree-BRT (Park and Kim 2019), quadratic discriminant analysis (Youssef and Pourghasemi 2021), meta-learning approach (Pradhan

et al. (2021), and generative adversarial networks (GAN) model (Al-Najjar et al. 2021). Dikshit et al. (2021) demonstrate the direction toward deep learning models in future studies for geohazard susceptibility assessment.

Landslide susceptibility assessment approaches have been developed over time. Dikshit et al. (2020) grouped the classification techniques used for landslide susceptibility modeling into four groups, namely qualitative, semi-quantitative, quantitative, and deterministic. In qualitative, mostly subjective methods, weights are assigned to each landslide causative factor based on the analyst's knowledge of the study area. Semi-quantitative methods of techniques such as the analytic hierarchy process (AHP) and the dangerous pixel approach are logical tools and focus on significant factors in attributing weights or values.

Analysis using quantitative methods, which include bivariate and multivariate models, depends on the density of landslides in the classes of each conditioning factor (Kumar and Anbalagan 2015). Bivariate statistical approaches such as frequency ratio, information value, and weights of evidence calculate the weight. However, multivariate models such as logistic regression take into consideration the collective effect of the parameters (Mandal and Mandal 2018). Deterministic approaches are based on the determination of the factor of safety (FS) through analysis of the soil's physical and mechanical properties. FS is the ratio between the parameters that favor the occurrence of landslides and those that contribute to the stabilization of the slopes (Kanungo et al. 2008).

Recently, researchers prefer the use of models based on machine learning (ML) algorithms, which have proven to be more performing in order to successfully overcome the limitations of other statistical techniques (Pham et al. 2016). The goal of our research is to assess and compare the performance of three different models: random forest (RF), decision tree (DT), and the artificial neural network (ANN), which is a machine learning-based algorithm. This comprehension allows us to draw more solid conclusions on the predictive capacity of the different models used. This research also aims to produce an accurate landslide susceptibility map for the N'fis watershed located in the High Atlas (Morocco) that will be useful for landslide risk mitigation and contribute to the sustainable development of the region. In addition, the results of this study are a valuable aid for local decision-makers in sustainable land management and also provide knowledge to understand the relationship between geo-environmental variables and landslide distribution. The principal novelty of the present study is that it provides an application and comparison, one to another, of the three models DT, RF, and ANN performance in landslide susceptibility assessment. We have identified and ranked the most important factors responsible for landslides in the study area, and we have evaluated the reliability and predictive power of

the three machine learning models based on sensitivity specificity, precision, accuracy, kappa, and area under the ROC curve (AUC). To the best of our knowledge, no similar study has been undertaken in the study area which is increasingly suffering from the severity of these phenomena except the one by Amaya et al. (2014) which evaluates ground instability events in this watershed using a traditional qualitative method. Hence the importance and necessity of such a study as the one undertaken in our research. We believe that the present research results could make a significant contribution to the scientific community.

Study area and data used

Study area

The N'fis is one of the most important sub-basins of the Tensift River. Geographically, it is located between latitudes 30.5° and 31.2°N and longitudes 7.55° and 8.40°W. Its surface area is about 1712 Km². The minimum altitude of the basin is order 669 m, while the highest point is 4143 m (Fig. 1).

The wadi N'fis drains this watershed from the south to the north of the region. Two barrages were constructed, contributing to the modification of the hydrodynamic and sedimentary behavior of the watershed (Gourfi et al. 2020). The Takerkoust dam and the Ouirgane dam were built in 1935 and 2007, respectively. Climatically, the N'fis area is located in an arid to the semi-arid region with an average

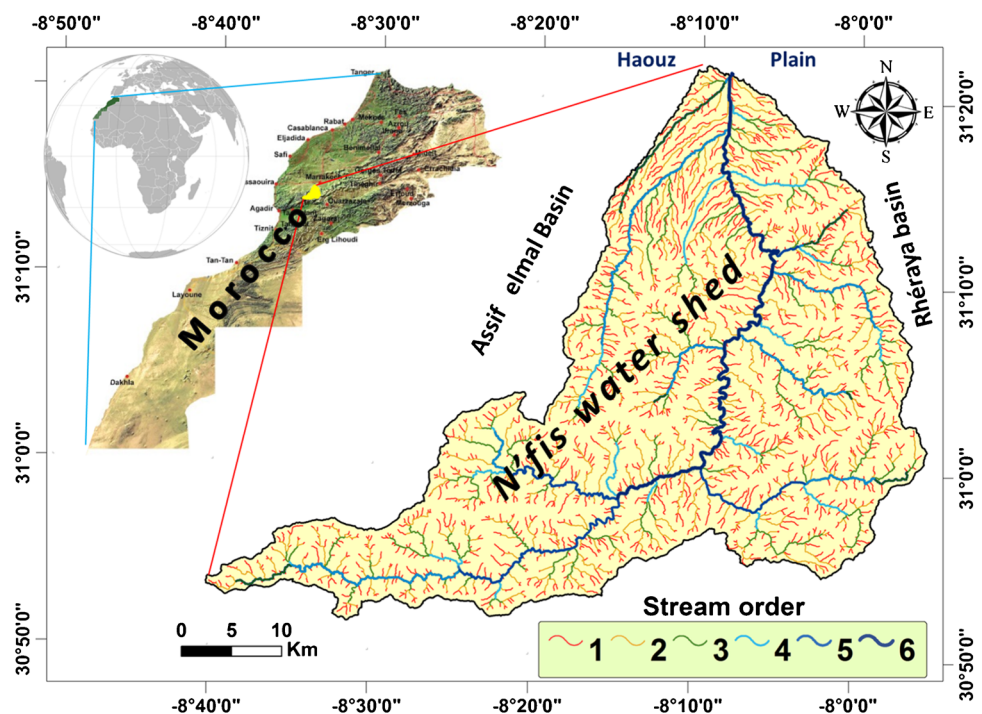
annual rainfall of 375 mm. The spatial distribution of these precipitations is not quite equal over the whole basin, in fact, they are more important in the south and southeast where the relief is higher. The low precipitation is measured in July and August, while the maximum values are recorded in March and April (Gourfi et al. 2020). There are two main seasons: a wet season from October to April, where practically all rainy events occur, accounting for an average of 84 percent of annual precipitation, and a dry season from May to September, representing just 16% of annual rainfall (Amaya et al. 2014). The average annual temperature is about 18.6 °C, with a minimum of 7.5 °C in January and a maximum of 47.5 °C in July.

Geologically the study area is part of the western basin of the High Atlas Mountains. The terrain shows several lithological facies whose age ranges from Paleozoic to Quaternary (Michard et al. 2008). In the southern part of the basin, we see a dominance of primary age rocks dominated by schistous formations in association with carbonate rocks and other magmatic rocks, as well as sandstones alternating with Permo-Triassic clays (Hollard et al. 1985). While the northern part of the basin is occupied by marl and limestone formations of the Upper Cretaceous and Plio-Quaternary.

Landslide inventory map

A crucial stage in landslide susceptibility modeling is the creation of a landslide inventory map. In this study, landslide data was obtained through an extensive field survey with handheld GPS devices and interpretations of Google

Fig. 1 Location of the study area



Earth images made in Google Earth Pro 7.3. A total of 156 landslide locations were identified. Stable areas were also determined based on several field missions during which we located stable sites in relation to landslides. For areas that are difficult to access, we referred to high-resolution satellite images based on our good knowledge of the study area. We have mapped many sites in order to cover as much of the N'fis basin as possible, and then we randomly selected the number of points requested in the modeling process.

In landslide modeling, there is no specific predefined method to categorize the inventory data. It is usually decided based on the accessibility and quality of the data (Youssef and Pourghasemi 2021). According to the literature, the percentages commonly applied to divide the inventory dataset are 70 and 30%, respectively, for the training and validation datasets (Pourghasemi and Rahmati 2018; Hong et al. 2016).

Adopting this sampling strategy, we randomly selected 109 (70%) of the landslide locations for the training phase of the models, and the remaining 47 (30%) landslides for the validation phase. As a result, an equal number of non-landslide pixels from the free-landslide area were randomly picked, with landslide pixels given a value of 1 and non-landslide pixels given a value of 0. Finally, the landslide inventory map was transformed to raster data with a 12.5-m grid size (Fig. 2).

Data collection

Several data sources were used in this research for the preparation of thematic maps of factors influencing landslide susceptibility. The Table 1 presents informations on the different supports used. The geological map and the soil map

Fig. 2 Landslide inventory map used in this study

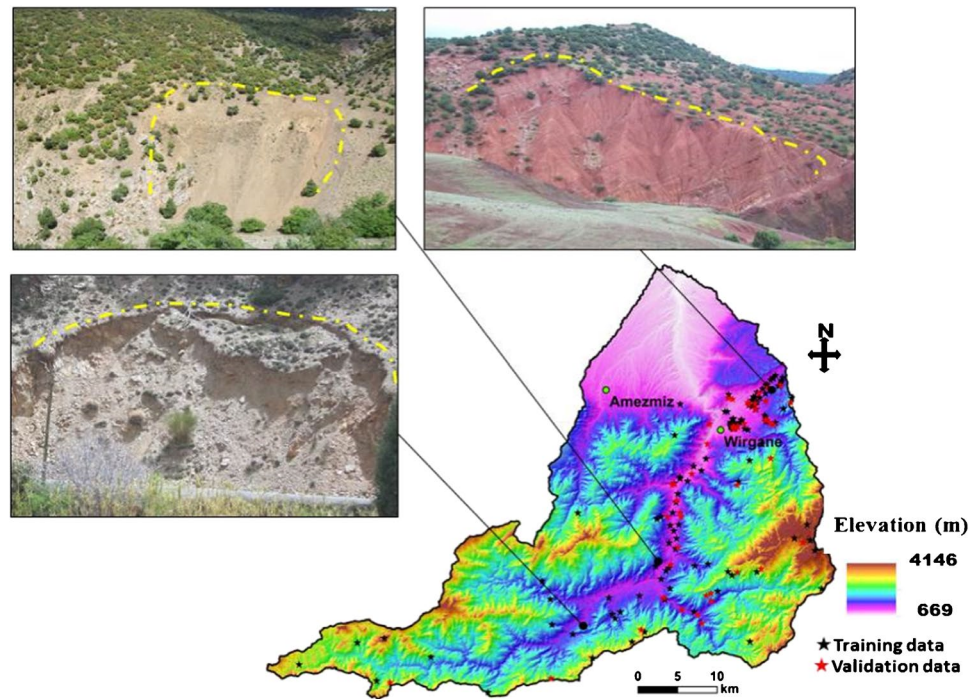


Table 1 Data sources used in our study

Data source		Resolution/scale	Date of acquisition
DEM downloaded from https://vertex.daac.asf.alaska.edu		12.5 m	2018
High resolution satellite image downloaded from the site: https://scihub.copernicus.eu	B02—B03—B04—B08	10 m	2019
	B05—B06—B07 B12—B08A—B11	20 m	
	B01—B09—B10	60 m	
Soil map (Mathieu. 2002)			2002
Geologic map (Marrakech 1/500,000)		1/500,000	
Rainfall data (ABHT. Water and forestry service)		-	2018

were digitized and cut out, along with the other supports, according to the territory of the N'fis watershed.

Landslide conditioning factors (LCFs)

Landslides are caused by a variety of reasons called landslide susceptibility factors (Sandric et al. 2019). The factors must be chosen according to the local features of the research region, data availability, landslide type, and the technique utilized in order to properly apply the approach and obtain a credible landslide susceptibility map (Jaafari et al. 2019). In this research, taking into consideration the most representative local morphological and spatial features, 14 causative factors were selected (Fig. 3). All data used in the present study were georeferenced in the north of Morocco.

Slope angle is the most important intrinsic element affecting slope stability (Hadji et al. 2013). It is frequently used in the preparation of landslide susceptibility maps. The slope angle map of the N'fis basin was generated based on a digital elevation model (DEM). It was then reclassified into nine classes, 0–5, 5–10, 10–15, 15–20, 20–25, 25–30, 30–35, 35–40, and 40–80° (Fig. 3A).

One of the most important elements influencing the distribution of slope stresses is altitude (Althuwaynee et al. 2016). The altitude at the level of the N'fis watershed varies from 610 to 4164 m. These altitude values have been divided into nine classes with an interval of 400 m (Fig. 3B).

The aspect determines the direction of the maximum slope of the land surface (Fang et al. 2020). Several researchers have considered the slope aspect as a determining factor in landslides (Meinhardt et al. 2015; Song et al. 2019). The slope aspect of the study area is classified as flat (1), north (337.5–360°, 0–22.5°), northeast (22.5–67.5°), east (67.5–112.5°), southeast (112.5–157.5°), south (157.5–202.5°), southwest (202.5–247.5°), west (247.5–292.5°), and northwest (292.5–337.5°) (Fig. 3C).

The rate of change of the slope is expressed by the profile curvature. As a result, it has an impact on the speed at which water drains from the surface, causing erosion and sediment movement below (Chen et al. 2016a, b). The profile curvature of the study area was also created from a DEM in ArcGIS 10.4 software and divided into three classes: < −0.05, −0.05–0.05, and > 0.05, respectively (Fig. 3D).

Distance to roads can be considered one of the primary anthropogenic variables impacting landslide incidence, as it can result in the generation of slopes cut by road construction, disrupting the natural topography and damaging slope stability (Al-Najjar and Pradhan 2021). In this study, the N'fis basin was divided into six buffer zones from the road network 0–100 m, 100–200, 200–300, 300–400, 400–500, and > 500 m (Fig. 3E).

The distance to rivers plays a very important role in the formation of landslides (Fang et al. 2021). Rivers are the main mechanisms contributing to the occurrence of landslides in mountainous regions (Park et al. 2013). In this research, seven distinct buffer zones were defined in the study area to evaluate the to which the streams affected the slopes (Fig. 3F).

Lithology is used by several researchers in many evaluations of landslide susceptibility. The different lithological characteristics of geological formations result in different levels of vulnerability to landslides. The lithological map of the study area is prepared on the basis of the geological map of Morocco 1:500,000. The study area is covered with different types of lithological units, as shown in (Fig. 3G).

Generally, precipitations can cause serious threats to people and their property worldwide. They may also decrease the factor of safety of slopes and trigger landslides (Grelle et al. 2014). This is because the suction of the soil material controls the shear strength of unsaturated soil (Santoso et al. 2011). During the infiltration of precipitation water, the suction of the material progressively decreases and thus leads to a reduction of the shear strength (Capparelli and Tiranti 2010). From the precipitation database provided by the Tensift Hydraulic Basin Agency, we created a precipitation map by interpolating annual mean precipitation at different stations covering the N'fis watershed. The resulting map has been reclassified into eight classes (Fig. 3H). This map shows that the greatest rains are in the eastern part of the basin as a result of the orographic effect near the Toubkal summit.

The normalized difference vegetation index (NDVI) is a way of measuring the amount of vegetation cover in a particular area (Jia et al. 2014). The current study's NDVI map, as illustrated in figure (Fig. 3I), was created using Sentinel-2 data from November 2019 using the following equation:

$$\frac{IR-R}{IR+R} \quad (1)$$

where R and IR are spectral reflectance values in the visible (red) and near-infrared, respectively.

Land use shows the land cover of an area relatively related to human activities. In general, areas with low vegetation density are more vulnerable to landslides than areas with dense vegetation (Yusof and Pradhan 2014). We have chosen to classify the N'fis watershed area into five land-use domains (Fig. 3J).

Faulting is one factor to be considered in mapping susceptibility to ground motion. Fracture and shear rates play an important role in determining the level of slope instability in highly faulted areas (Shahabi et al. 2014). In general, it is observed that several landslide events have occurred at sites close to the lineaments. In this study, seven different buffer zones were created around existing faults (Fig. 3K).

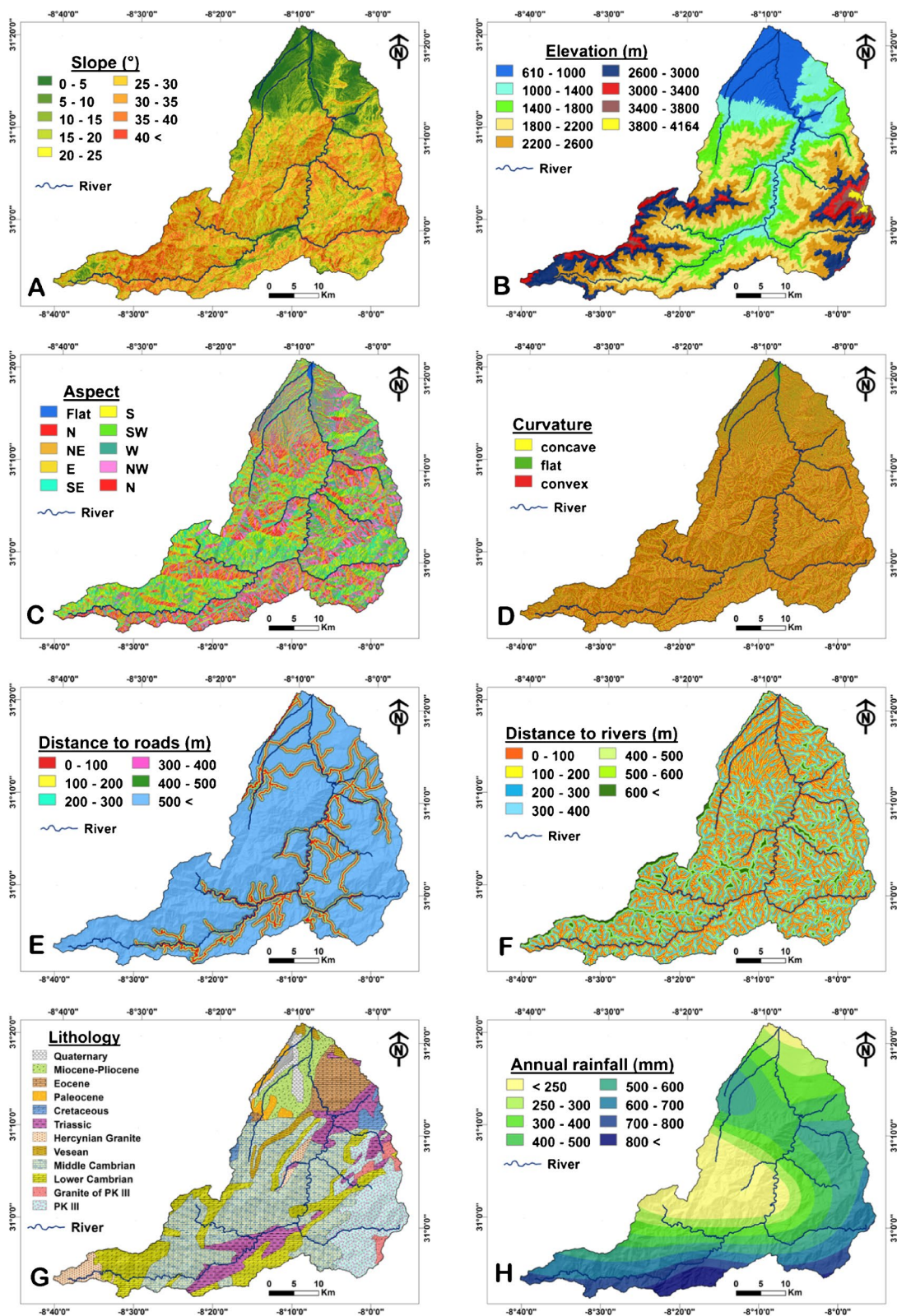


Fig. 3 Landslide conditioning factors: **A** slope, **B** elevation, **C** aspect, **D** profile curvature, **E** distance to roads, **F** distance to rivers, **G** lithology, **H** Precipitation, **I** NDVI, **J** land use, **K** distance to faults, **L** TWI, **M** soil type, and **N** TPI

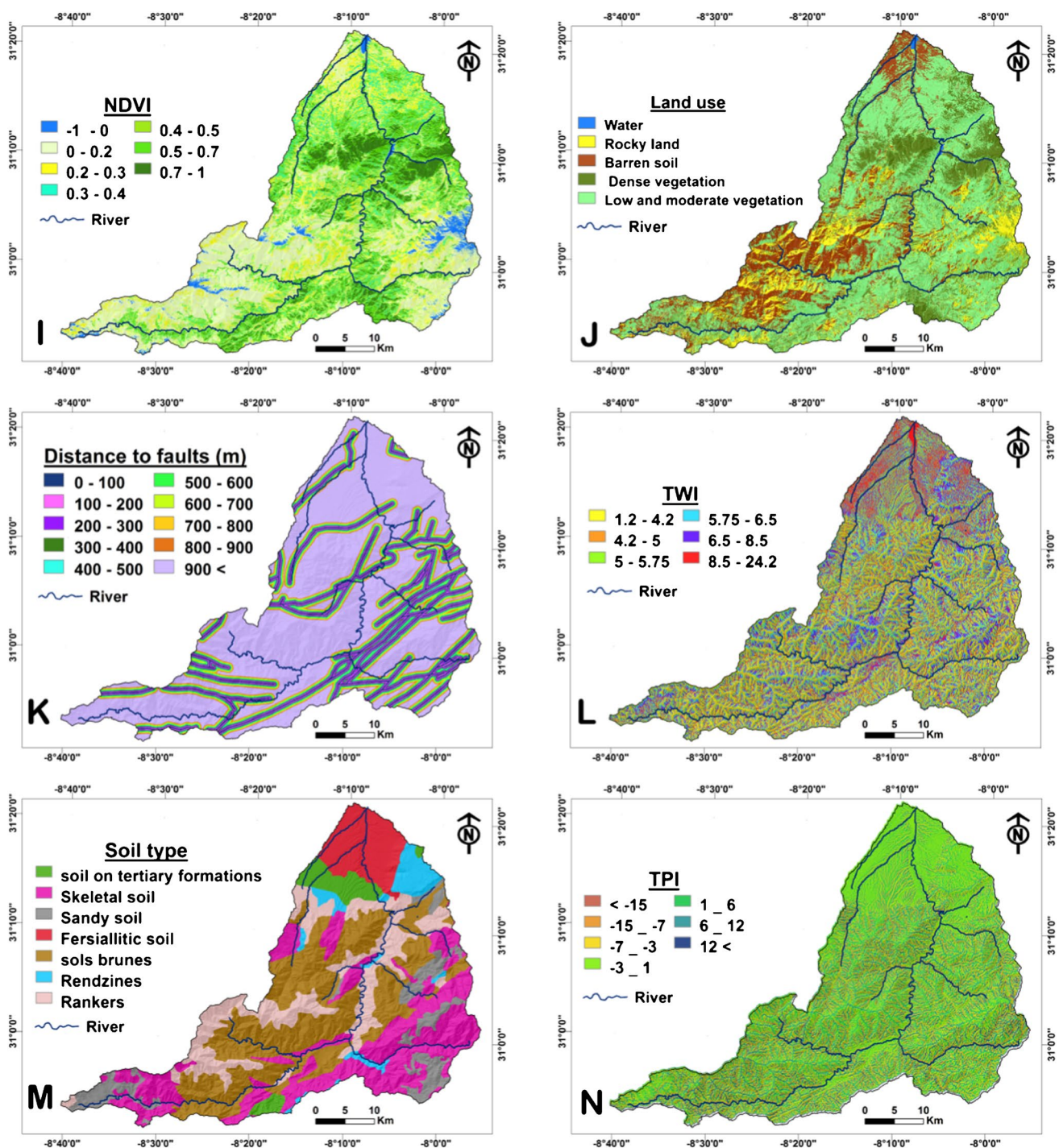


Fig. 3 (continued)

The topographic wetness index (TWI) is one of the primary topographic parameters in the streaming model (Pourghasemi et al. 2013). In this research, the TWI values, obtained based on a DEM covering the N'fis region, were classified into seven classes: -97 to -15 ; -15 to -7 ; -7 to -3 ; -3 to 1 ; 1 to 6 ; 6 to 12 , and 12 to 104 (Fig. 3L).

The soil type has a direct impact on the processes of infiltration and evaporation and therefore humidity on the surface of the earth. It also influences the type and rate of vegetation that can impact the instability of the landscape. The distribution of soils in the N'fis basin, illustrated by the map in Figure (Fig. 3M), depends primarily on the nature

of the bedrock, which is a function of altitude, slope, and exposure of the slopes (Merkhi et al. 2015).

The topographic position index (TPI) reflects the difference between the cell's altitude and that of the neighboring cells (Yusof and Pradhan 2014). Positive TPI values represent locations above the average of their surroundings as defined by the neighborhood (ridges). Terrains with a constant slope or flat areas have TPI values close to zero. While the negative TPI values indicate sites that are located below their environment, such as valleys. The N's Bassin shows TPI values that range from -15 to over 12 indicating that the watershed is steep (Fig. 3N).

Methodology

The assessment of landslide susceptibility in the N'fis watershed, which is the subject of this study, was carried out in a four step procedure: (a) constructing a dataset; (b) elaboration of training and validation datasets from a landslide inventory and choosing predisposing factors; (c) generation

of landslide susceptibility maps; (d) verification and comparison of the approaches that have been used (Fig. 4).

Landslide dataset preparation

In the current study, the frequency ratio method was used to examine correlations between landslides and their causative factors. The frequency ratios are derived by dividing the landslide occurrence ratio for each factor's class by an area ratio for each factor's class compared to the total area. The obtained FRs were normalized from 0 to 1 for each landslide factor. These datasets were then utilized to create the training and validation datasets, which were used to run the models and assess their predictive ability.

Selection of landslide factors

The information gain (IG) method was used to infer the importance of each factor. An elevated GI index for a causative factor indicates that it is more essential for landslide models. On the other hand, a factor with a GI value of zero should be removed from the data set because it does not

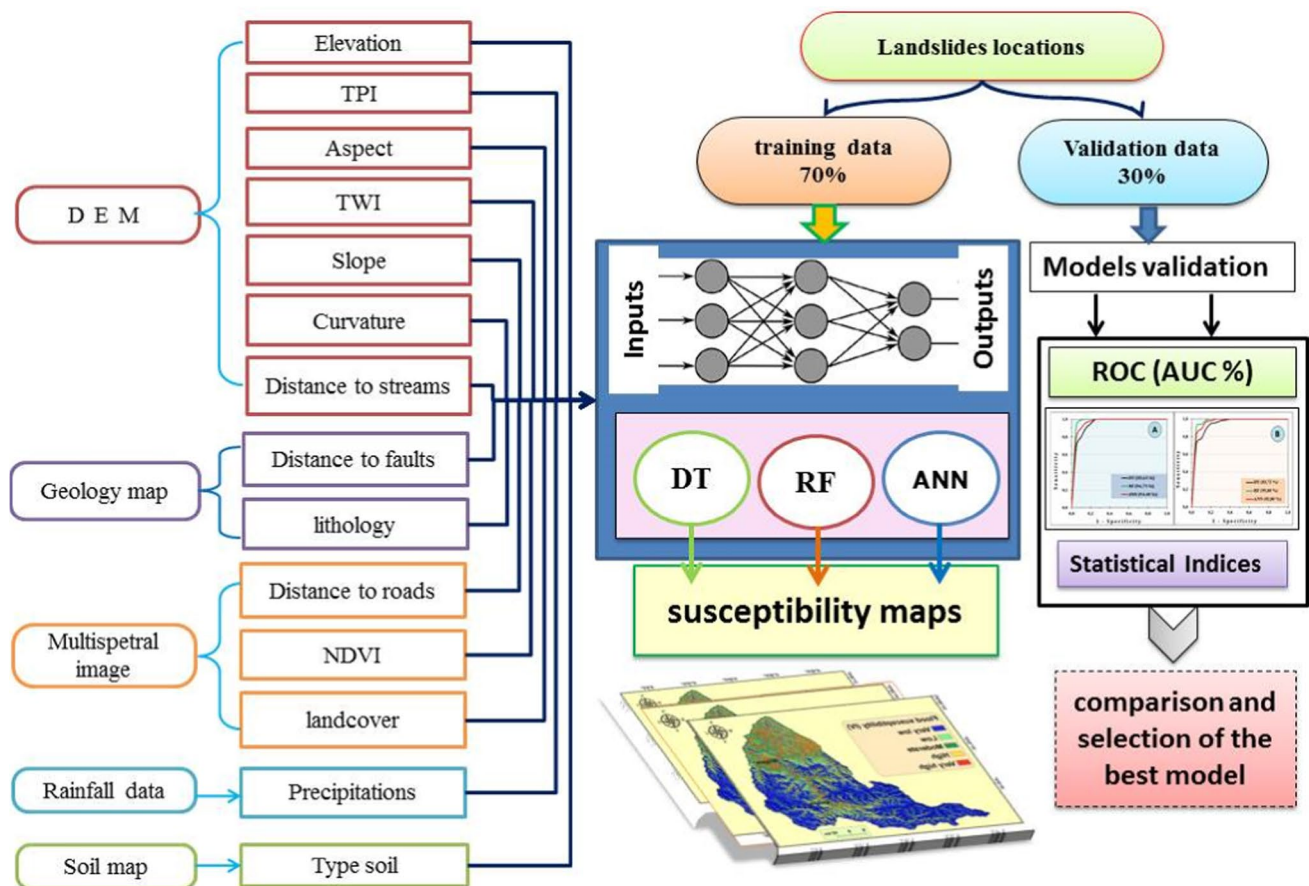


Fig. 4 Flow chart illustrating the modeling strategy used in this study

contribute (Park and Kim 2019). For all of the landslide causal elements, Table 2 displays the information gain index and associated standard deviation. Because the IGA results calculated for such factors were more than 0, it is clear that all of the parameters employed in this study should be included in the landslide susceptibility calculations. The estimation of the contribution of the causative factors shows that the slope factor ($Am=0.098$) has the most important contribution to the occurrence of landslides. In fact, as the slope angle increases, the probability of landslide occurrence also increases. This can be explained by the fact that as the slope angle increases, the driving force of the slope mass also increases, while the resistive force decreases. Human activities such as road construction and enlargements play an important role in the initiation of landslides. These activities generally require cutting or excavating slopes, which changes the original geological conditions and subsequently disrupts the stability of the slopes. For this reason, it is important to consider the distance to roads as a causative factor for landslides when modeling. The occurrence of landslides was also controlled by the distance to rivers (with $AM=0.069$ in our study). Active incision of a river or stream has a significant influence on slope instability through toe erosion and saturation of slope materials by causing additional slope steepness and instability.

Modeling approach based on machine learning models

Random forest

Random forest (RF) was developed by Breiman in 2001 (Park et al. 2019). It is a method of model construction that

Table 2 Information gain values for the landslide factors used in this study

Landslide factors	Average merit	Standard deviation
Slope	0.098	0.0310
Distance to roads	0.07	0.0282
Distance to rivers	0.069	0.0281
Lithology	0.054	0.0273
Altitude	0.051	0.0272
Precipitation	0.048	0.0271
Soil type	0.046	0.0271
TWI	0.032	0.0271
Land use	0.029	0.0272
Aspect	0.018	0.0276
TPI	0.016	0.0277
Distance to faults	0.009	0.0281
NDVI	0.005	0.0285
Curvature	0.004	0.0285

gives estimators of the Bayes classifier (Hong et al. 2016). It employs a number of tree predictors, each of which is dependent on the values of vectors picked at random and spread uniformly over the forest's trees (Chen et al. 2018a, b). Several studies have indicated that a set of classification trees is more precise than each of its individual members (Youssef and Pourghasemi 2021). Hong et al. (2016) show that RF generates a variety of measures of variable significance, the most reliable of which is based on the reduced classification performance when the values of a variable in a tree node are swapped at random.

To obtain the best model performance and minimize the out-of-bag (OOB) error, two parameters must be optimized by the user: the number of variables tested at each node (mtry) and the number of trees in the forest (ntree).

Two types of error rates should be examined for a better categorization of parameters and selection of the most significant ones: the average decline in accuracy and the average fall in the Gini coefficient (Pourghasemi and Kerle, 2016, 2013).

There are n variables that may be picked as random subsets of the training data to find the suitable node to divide. The best distribution of nodes can be determined using Gini criteria (Chen et al. 2018a, b). This metric determines how closely variables and outcomes are linked. The random forest method considers the minimum mean to be the optimum distribution for each node (Kausar and Majid 2016). The Gini criteria are represented by the equation below (Chen et al. 2018a, b):

$$\text{Gini}(K, x_i) = \sum_{i=1}^m \frac{a_i}{n_s} I(K_{ui}) \quad (2)$$

where m represents the number of landslides at each node k and n_s is the number of training input feature vectors. $I(k_{ui})$ represents the distribution of class labels at a node. At node k , the value p is a feature variable $x_i \in X$ where $x_i = \{u_1, u_2, \dots, u_m\}$ and the value of $I(k_{ui})$ can be calculated as

$$I(K_{ui}) = 1 - \sum_{i=0}^c \frac{n_{ci}^2}{a_i^2} \quad (3)$$

where n_{ci} are the samples that belong to c_i with values u_i and a_i is the number of samples with the value u_i at node k .

Decision tree

A decision tree (DT) is a nonparametric approach for detecting nonlinear and nonadditive correlations between input parameters and targeted or predictive elements (Tsangaratos and Ilia 2016). It is a hierarchical model made up of internal nodes, leaf nodes, and branches that is represented by a tree-like structure. One of the advantages of this model is that it is easy to apply with easily interpretable results. The main

inconvenient of decision trees is that multiple outputs are not allowed, and this model is sensitive to noisy data (Zhao and Zhang 2008).

Decision tree classification can be carried out using several learning algorithms, like CART, Iterative Dichotomiser 3 (ID3) and C4.5. In all three models, the entities in the input dataset may be independent. However, they differ in how they measure differentiation and variety criteria (Tsangaratos and Ilia 2016). In our present study, the landslide inventory data is used as the dependent variable while the independent variables are represented by the landslide conditioning factors. For the data analysis, the C4.5 method was applied. The tree starts at a node, which is then divided by choosing the attribute that best categorizes a group of samples based on an attribute selection measure. The measure of attribute selection in the c4.5 model is based on the entropy concept which reflects the disorder degree. As a result, a tree develops by selecting the lowest entropy characteristic. The entropy at node N is calculated using the equation below (Yeon et al. 2010):

$$\text{Entropy}(n) = - \sum_j P(C_j/N) \log_2 P(C_j/N) \quad (4)$$

where $p(C_j|N)$ is the relative frequency of N . Of the k attributes of N , the entropy for selecting attribute A is given by:

$$\text{Entropy}_A(N) = \sum_{j=1}^K \frac{|N_j|}{|N|} \times \text{Entropy}(N_j) \quad (5)$$

InfoGain is a gain from the differences between the entropy of the original node and the entropy of the newly split nodes. The equation is as follows:

$$\text{Infogain}(A) = \text{Entropy}(A) - \text{Entropy}_A(N) \quad (6)$$

As a result, C4.5 selects the property with the smallest entropy or largest InfoGain. This feature makes the tree expand towards continuous continuous attributes. To solve this problem, the InfoGain is normalized by SplitInfo, a kind of entropy on the separation point of an attribute. Thus, it has a high value for an attribute with a certain number of splits. When node N is split into n subsets, the SplitInfo equation is as follows:

$$\text{Split Info} = - \sum_{i=1}^v \frac{|N_j|}{|N|} \times \log_2 \frac{|N_j|}{|N|} \quad (7)$$

thus, InfoGain compensated by SplitInfo is GainRatio, which is defined as follows:

$$\text{Gain Ratio}(A) = \frac{\text{Info Gain}(A)}{\text{Split Info}(A)} \quad (8)$$

Artificial neural network

The artificial neural network (ANN) algorithm was originally designed based on the multilayer perceptron with three neural layers: an input layer that receives the environmental input elements to be used in the ANN model's training, validation, testing, and application. The intermediate or hidden layer is devoted to the interactions between the input parameters and matches the input to the output, allowing the ANN to learn complex tasks and extract meaningful data. This allows the ANN to learn complex tasks and extract meaningful features from the input patterns and the output layer, which gives the result of the input processing (Bragagnolo et al. 2020).

The focus of ANN computing is to build a new model of the data generating process in such a way that outputs can be generalized and predicted from inputs (Kawabata and Bandibas 2009). In the training phase, the network will predict the label of each sample using its weights and bias values. A cost function finds the distinction between the computed label and the true label. In the backpropagation step, each weight receives an update based on the gradient of the cost function. This process continues until convergence is reached or a maximum number of epochs is reached in the learning.

One of the more interesting neural calculation methodologies is error back-propagation neural network (EBPNN) computing. In this study, a backpropagation algorithm was used. The inputs x_n , $n=1, \dots, n$ to the neuron are multiplied by weights w_{ni} and summed with the constant bias term Q_i . The result n_i obtained is the input to the activation function y . For mathematical convenience, the activation function is a hyperbolic tangent ($\tanh$) defined as follows (Yilmaz 2009):

$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (9)$$

The output of node i becomes.

$$y_i = f\left(\sum_{j=1}^k w_{ik}x_i + Q_i\right) \quad (10)$$

The backpropagation algorithm's mathematical representation is described by the equations below:

$$\text{net}_i = \sum_{\forall j, w_{ji} \neq 0} w_{ij} * a_j \quad (11)$$

$$a_i = \frac{1}{1 + e^{-\text{net}_i}} \quad (12)$$

$$E_p = \sum_{i \in \text{Output units}} (a_i - o_i)^2 \quad (13)$$

$$E = \sum_p E_p \quad (14)$$

Partial derivative for pattern p with respect to w_{ij} :

$$\frac{\partial E_p}{\partial w_{ij}} \quad (15)$$

Partial derivative for the whole training set with respect to w_{ij} :

$$\frac{\partial E}{\partial w_{ij}} \quad (16)$$

$$\frac{\partial E}{\partial w_{ij}} = \sum_p \frac{\partial E_p}{\partial w_{ij}} \quad (17)$$

$$\nabla E = \frac{\partial E}{\partial w_1}, \frac{\partial E}{\partial w_2}, \dots, \frac{\partial E}{\partial w_n} \quad (18)$$

$$\nabla E_p = \frac{\partial E_p}{\partial w_1}, \frac{\partial E_p}{\partial w_2}, \dots, \frac{\partial E_p}{\partial w_n} \quad (19)$$

$$w_{ij}(n+1) = \Delta w_{ij}(n) + w_{ij}(n) \quad (20)$$

In Eqs. 9–19; a_i : activation of unit i , w_{ij} : connection strength from unit i to unit j , o_i : desired activation of output unit i (i th's component of the output vector), net_i : net input to unit i , E_p : quadratic error for pattern p , E : total quadratic error on the training set, ∇E : gradient with respect to the whole training sets, ∇E_p : gradient with respect to pattern p , and $\Delta w_{ij}(n)$: weight update of w_{ij} in the n th learning step.

Model evaluation measures

Several statistical criteria were employed to assess the accuracy of the tree models utilized in this study. The probabilities provided by each model were compared to historical landslide locations to produce a confusion matrix that includes true negative (TN), false negative (FN), true positive (TP), false positive (FP), and values based on how well the landslide event was categorized (Park et al. 2019). The statistical indices calculated are

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (21)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (22)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (23)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (24)$$

Sensitivity is the portion of correctly classified landslide pixels, while specificity is the portion of correctly classified non-slide pixels. Accuracy a positive predictive value that represents the ratio of correctly classified landslide and non-slide pixels. A model is of high quality if its precision and accuracy are higher. A perfect model will have values equal to one (Witten and Frank 2002). In addition to these indices, the kappa coefficient is calculated for our models. This coefficient allows the evaluation of the reliability of each model (Garosi et al. 2019):

$$K = \frac{P_{obs} - P_{exp}}{1 - P_{exp}} \quad (25)$$

$$P_{obs} = \frac{TP + TN}{SA} \quad (26)$$

$$P_{exp} = \frac{(TP + FN)(TP + FP) + (FP + TN)(FN + TN)}{SA^2} \quad (27)$$

where SA is the total number of pixels in the study area. The value of the kappa coefficient varies between 0 and 1. If it is equal to or greater than 0.8, the performance of the model is considered almost perfect (Landis and Koch 1977). The ROC curve is the sensitivity versus specificity:

$$\text{AUROC} = \frac{\sum TP + \sum TN}{L + N} \quad (28)$$

where N is the total number of no-landslides and L is the total number of landslides.

Results and analysis

Spatial relationships and importance of variables

It is critical to assess the correlation of specified causal elements with previous landslides when predicting landslide susceptibility. This allows an adequate selection of factors to improve the performance of methods used for this modeling. We used the FR index to measure the spatial relationship between each causative factor and the landslides occurring in the N'fis watershed.

The results illustrated in Fig. 5 show that the terrain located at a distance of 100 m from the roads is the most vulnerable (FR = 4.23), which proves the effect of anthropic activities on the stability of the slopes. The Cambrian and Triassic lands (FR > 2) are also exposed to the risk of landslides because of their geological nature and the importance of the surface they occupy. The slope shows a good

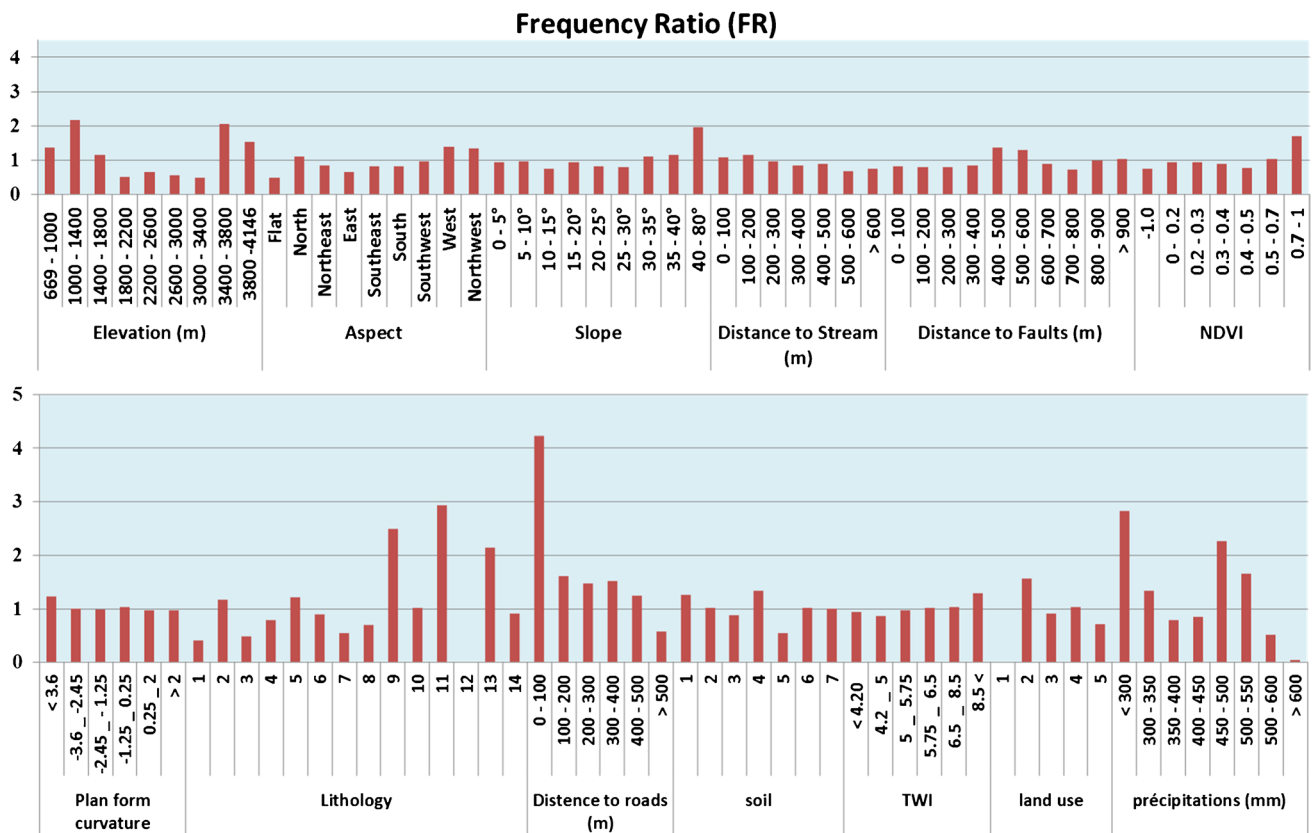


Fig. 5 The historical landslide frequency ratio in each class of conditioning factors

correlation with the landslide phenomenon. In fact, the more the slope dips increases, the more the susceptibility to the risk increases, and from 30° the FR index becomes higher than 1. Concerning the distance to the watercourse, the risk weakens as one moves away from the watercourse. The sites whose NDVI value is higher than 0.7 are also vulnerable, which can be expressed by the fact that most of the forests in the N'fis watershed are planted on steep slopes. For the aspect factor, we note that the risk is greater on the slopes oriented from north to west ($FR > 1$). Areas with an altitude of between 669 and 1800 m or above 3800 m are at risk of landslides, while those with an altitude range between 1800 and 3800 m are relatively more stable.

Landslide susceptibility mapping

Landslide susceptibility maps for the N'fis watershed were created in two steps in this study. In the first step, the DT, RF, and ANN models were trained using the R programming language and the training dataset. These three models were applied after a tenfold cross-validation to reduce variability. To enhance the predictive performance of the landslide models, the values of the parameters for each model were optimized during the training phase. These

models were then used to generate landslide susceptibility indices (LSI) for our study area. The second step is reserved for the generation and reclassification of the obtained maps in a GIS using the natural breaks method into five susceptibility levels, namely very low, low, moderate, high, and very high.

Decision trees

We have used the package “rpart” to build our landslide model following the DT method. The package “mlr” was used to optimize the essential parameters: minsplit, minbucket, maxdepth, and cp. Maxdepth is the maximum depth of a tree, The minimal number of measurements in the parent node is called minsplit, while minbucket is the minimum number of observations in a terminal node, and cp represents the complexity parameter. The optimized values are 6 for minsplit, 6 for minbucket, 7.08 for maxdepth, and 0.002 for cp. To create a susceptibility map, the LSI values were classified into very low susceptibility (0.00–0.09), low susceptibility (0.09–0.30), moderate susceptibility (0.30–0.49), high susceptibility (0.49–0.70), and very high susceptibility (0.70–1.00) (Fig. 7A).

Random forest

Concerning the RF model, the hyper-parameters *mtry* and *ntree*, which reflect the number of variables evaluated at each node and the number of trees in the forest, were selected using the “caret” package. The ideal choices for *mtry* are 500 and 5 for *ntree*. The model construction was implemented using the “randomForest” package.

After model construction, the importance values of the 14 factors were calculated. Figure 6 presents the results obtained from this analysis. They show that all these factors have equally significant contributions to the occurrence of landslides. The slope factor had the highest significance value (0.251), followed by the distance to the road (0.149) and the distance to the rivers (0.142). The minimum value was attributed to the curvature factor (0.011).

To generate the susceptibility map, we have reclassified the index obtained into five susceptibility classes (Fig. 7B): very low (0.00–0.25), low (0.25–0.46), moderate (0.46–0.62), high (0.62–0.83), and very high (0.83–1.00).

Artificial neural network

The ANN model was constructed and trained with the following parameters: 300 learning epochs, the learning rate is 0.03, the maximum iteration is 1100, and the activation function is sigmoid. The hidden layer has a minimum and a maximum number of neurons of 2 and 12, respectively. The indices obtained according to the ANN model were also reclassified into five susceptibility levels (Fig. 7C): very low (0.00–0.27), low (0.27–0.39), moderate (0.39–0.68), high (0.68–0.85), and very high (0.85–1.00).

The final map shows sites sensitive to landslides well represented by the three models applied, presenting overall a similar spatial pattern, emphasizing the areas located in the proximity of rivers and on the slopes of steep slopes in the southeast of the N’fis basin. Alternatively, the regions of low to very low susceptibility are located in the center of the basin at the level of the Ijoukak region, formed by a depression with a low inclination, and at the northern end of the basin, which constitutes a transition zone between the high mountains and the Haouz plain. However, it is essential to note that some areas with significant differences have been identified in the maps, which demonstrates the uncertainties inherent in the used cartographic methods.

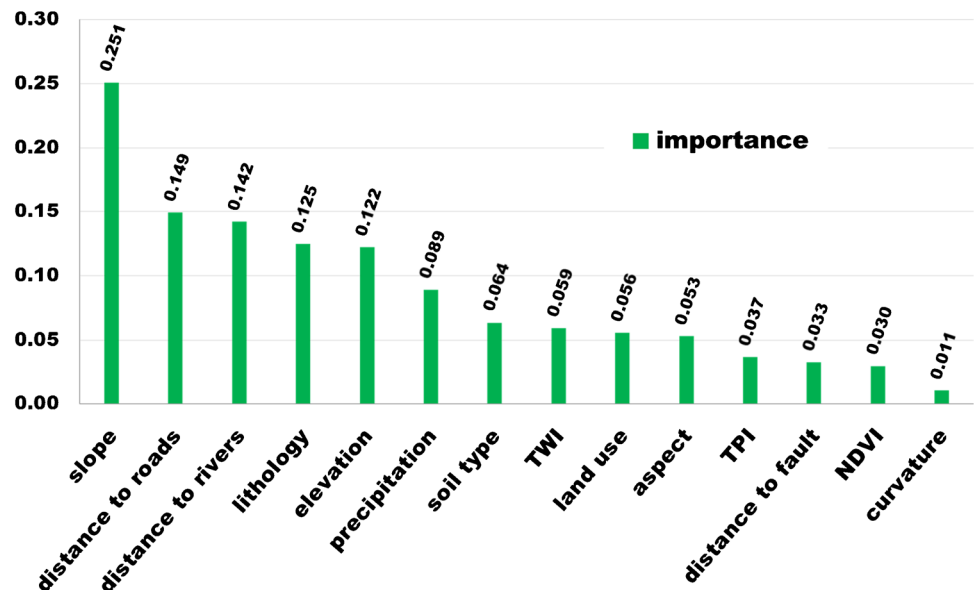
Figure 8 illustrates graphically the percentages of the surface of each class. On average, the three models show that 21.08% of the area of the N’fis basin is very susceptible to landslides, 27.29% is susceptible, 19.5% is moderately susceptible, while 17.72 and 14.41% have low and very low susceptibility, respectively.

Validation and comparison of models

Statistical indices

Several statistical indices are used to evaluate the performance of the three models used in this study. The results are presented in Table 3. In summary, the RF model outperformed the others since it had the highest sensitivity values (0.999), specificity (0.990), precision (0.997), and accuracy (0.987). The performance indices obtained for the ANN model are slightly lower than those offered by the RF model. In our study, the DT model did not perform well, with the lowest performance indices. The values of the kappa index

Fig. 6 Relative importance of conditioning factors in this study



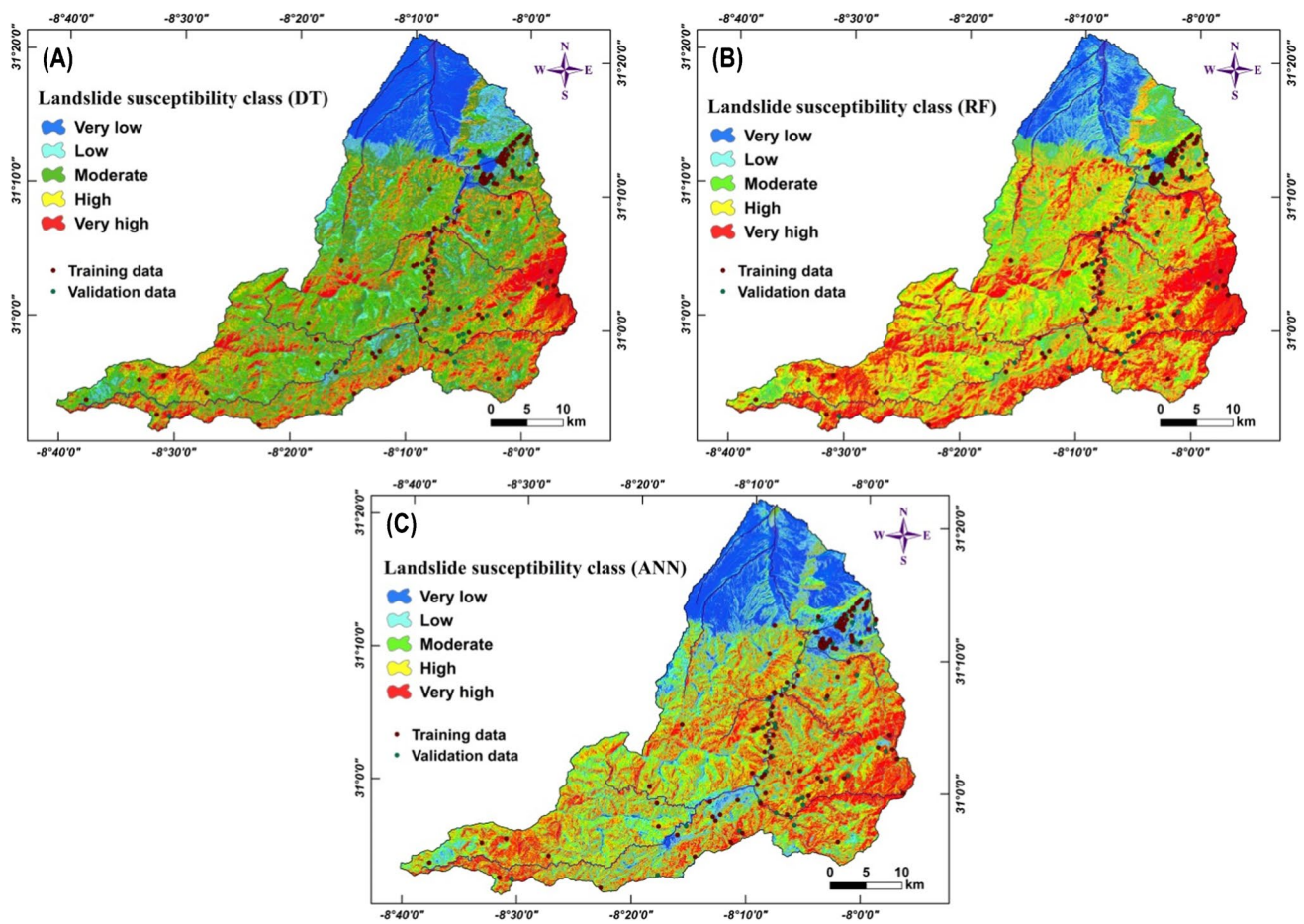


Fig. 7 Landslide susceptibility maps produced by: A DT, B RF, and C ANN models

Fig. 8 Distribution of each susceptibility class on LSMs

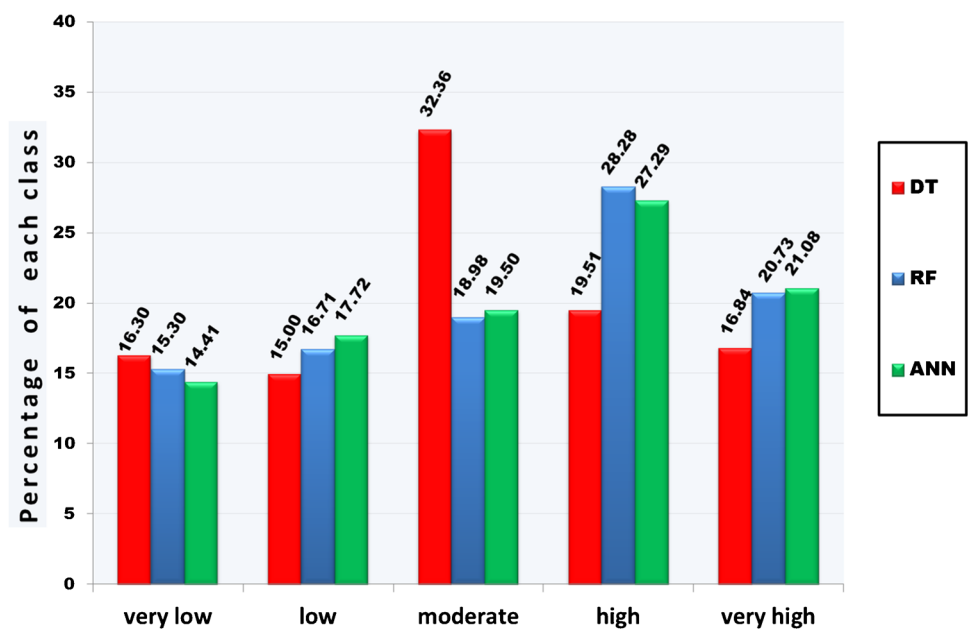


Table 3 Performance assessment of the three models with the training and validation datasets using statistical indices

	Training dataset			Validation dataset		
	DT	RF	ANN	DT	RF	ANN
Sensitivity	0.961	0.999	0.996	0.946	0.993	0.993
Specificity	0.963	0.990	0.971	0.958	0.974	0.951
Precision	0.963	0.997	0.993	0.959	0.994	0.988
Accuracy	0.962	0.997	0.992	0.952	0.990	0.985
Kappa	0.730	0.922	0.894	0.616	0.947	0.883

of the three models applied in this research vary from 0.730 to 0.922, revealing an overall similarity.

The performance of the three models was also evaluated with the validation data set. The RF landslide model is still the best-performing model. Followed by the ANN model and the DT model, the worst results were again given. The kappa index of the RF and ANN models is 0.947 and 0.883, respectively, which indicates a concordance of the results. For the DT model, the kappa index, on the other hand, was lower (0.616).

ROC method

The values of the area under the curve (AUC) calculated show that the best performance is obtained by the RF model (96.75%), followed by the ANN model (94.40%), while for the DT model an AUC equal to (85.65%) was obtained. Similarly, the predictive capacity of the three models, based on the use of the validation data set, follows the same ranking, with the RF model having the highest value (95.68%), followed by ANN (92.89%) and DT (83.72%). These results are in agreement with those obtained using the statistical

indices and indicate that the RF model is the best predictor of landslides among the three models (Fig. 9).

Discussion

This research is an application of three models, based on machine learning methods, to assess landslide susceptibility in the N'fis watershed. Several causative factors (slope angle, distance to faults, lithology, slope aspect, land use, distance to rivers, curvature, distance to roads, elevation, topographic position index (TPI), topographic wetness index (TWI), precipitation, soil type, and normalized vegetation index (NDVI)) were selected and crossed with data from a historical landslides inventory. From the frequency ratio calculations, it can be concluded that all selected factors can influence the landslide occurrence. The models' performance was evaluated by popular methods based on both the training and validation data sets. These validation techniques indicated that, overall, all three models performed well.

The DT model is an easily applicable and directly interpretable method. It also has several advantages: (1) It is a type of statistical analysis that does not make use of the

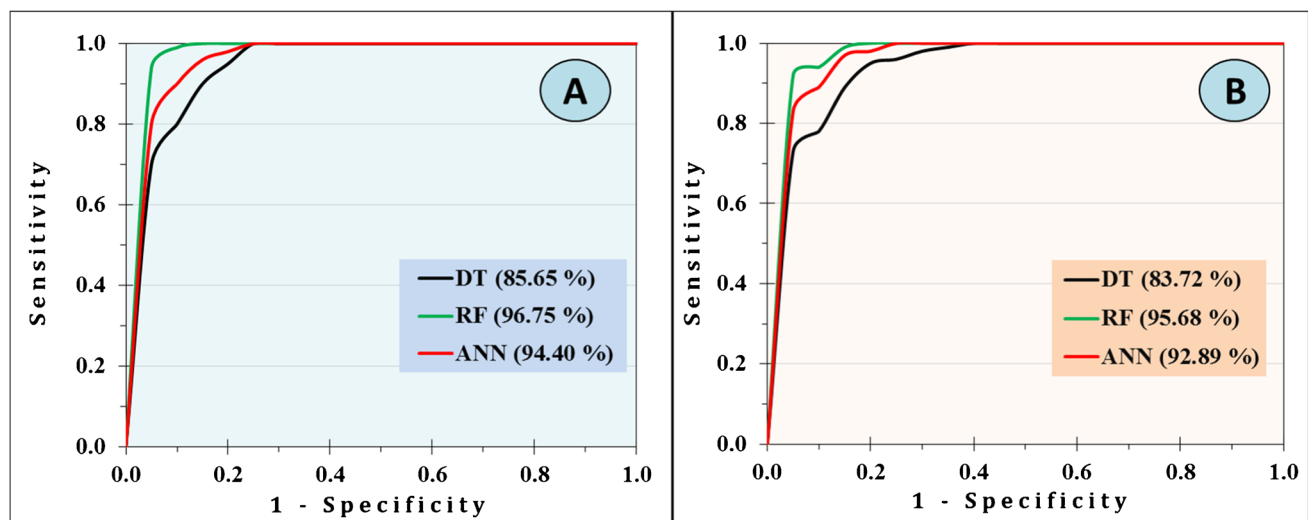


Fig. 9 ROC curves and AUC values of models in the **A** learning and **B** validation phases

assumption of a statistical distribution, (2) it can handle data from different measurement scales, and (3) it makes it easier to construct prediction rules for complex relationships. Despite these advantages, the DT model is too basic to fully describe many real-world situations. This is evident from the study's findings, which show that the LSM created by the DT model performed the worst in terms of statistical index analysis and area under the ROC curve.

The RF model, on the other hand, outperformed the ANN and DT models because it improved fit quality and predictive ability. This is reasonable since, as compared to basic classifiers, ensemble learning-based classifiers can minimize both bias and variance while avoiding overfitting issues. These findings are consistent with recent research demonstrating that ensemble learning aids in the improvement of simple classifier performance (Zhang et al. 2017).

According to the results obtained in this study, three factors have a significant contribution to the generation of landslides: The slope factor operates by the permanent effect of gravity, which promotes the displacement of unstable ground on the slopes. The greater the slope, the greater the susceptibility to landslides. Excavation during construction or widening of road networks also destabilizes slopes. For this reason, areas around roads are often prone to landslides. The distance to the river is also an important factor that can contribute to the occurrence of landslides. Indeed, the erosion of the banks threatens the stability of the slopes near the watercourses, which can give rise to landslides.

Some of the limitations of this study include the absence of some effective data, such as soil depth and texture, and distance to the water table. It is recommended to use these parameters in future research to enhance the predictive capability of the models, to generate more accurate landslide susceptibility maps, and to correlate this new type of geo-environmental factor to landslide occurrence. A hybridization of learning algorithms could potentially increase the algorithm's model parameters and predictive capability. We also urge that future studies include an effective investigation of the no landslide surface ratio, which was chosen to be the same as the landslide surface area despite the fact that the true no landslide area was much larger than the landslide zone. In addition, the sampling ratio (70/30) for both the training and validation data sets was evaluated. Finally, it is strongly recommended to focus on deep learning models capable of best deducing complex spatio-temporal relationships between variables. As suggested by Dikshit et al. (2021), the coupling of data-based models with physical models can also help in a better understanding of landslide processes and obtain, therefore explainable and interpretable machine learning models.

Conclusion

The development of an accurate landslide susceptibility map is a critical step in reducing the effects of these disasters. It is an indispensable tool for sustainable land use planning and management. In this research, we tried to map the areas susceptible to landslides in the N'fis watershed located in the High Atlas of Marrakech (Morocco). The objective of this study was to evaluate and compare the spatial prediction of landslides by three machine learning models using a landslide inventory prepared on the basis of data collected from several field missions and satellite image analysis.

Fourteen factors influencing the occurrence of landslides were analyzed and considered in this study: elevation, slope angle, slope aspect, curvature, lithology, land use, distance to faults, distance to rivers and roads, TPI, TWI, precipitation, soil type, and NDVI. The thematic maps of these causative factors were crossed with an inventory of historical landslides, of which 70% of the total data was reserved for model training and the rest for estimation of their predictive capabilities. The results obtained by the DT, RF, and ANN models were evaluated by several statistical indices and with the estimation of the area under the ROC curve. Overall, the three models showed good performances with the training and validation data sets. The analysis of the performance indices shows that the RF model performed the best, followed by ANN, while the DT model had the lowest indices. On average, the final maps estimate that about 19.25% of the study area is placed in the very high landslide sensitivity class. This figure is an alarm for land-use managers and planners to take the necessary measures to protect human communities from these devastating phenomena.

Since the prediction of landslides remains a difficult task and requires extensive field measurements, the performance of the models applied in this study proves that they can be used as interesting alternative tools to obtain reliable estimates of landslide susceptibility at a convenient computation time. These models can also be applied to various environmental problems such as floods, land subsidence, and soil degradation by erosion.

Finally, it can be concluded that the maps developed in this research will be beneficial to authorities for landslide risk mitigation and land use planning in the study area and other places with similar environmental contexts.

Declarations

Conflict of interest The authors declare that they have no competing interests.

References

- Al-Najjar HH, Pradhan B (2021) Spatial landslide susceptibility assessment using machine learning techniques assisted by additional data created with generative adversarial networks. *Geosci Front* 12(2):625–637
- Althuwaynee OF, Pradhan B, Lee S (2016) A novel integrated model for assessing landslide susceptibility mapping using CHAID and AHP pair-wise comparison. *Int J Remote Sens* 37(5):1190–1209
- Amaya A, Algouti A, Algouti A, El Aaggad N (2014) Mapping of mass movements hazard in the N'fis watershed, High Atlas, Morocco. *Int J Innov Appl Stud* 8(2):645
- Bragagnolo L, da Silva RV, Grzybowski JMV (2020) Landslide susceptibility mapping with r landslide: a free open-source GIS-integrated tool based on artificial neural networks. *Environ Modell Software* 123:104565
- Capparelli G, Tiranti D (2010) Application of the MoniFLaIR early warning system for rainfall-induced landslides in Piedmont region (Italy). *Landslides* 7(4):401–410
- Chen T, Niu R, Jia X (2016a) A comparison of information value and logistic regression models in landslide susceptibility mapping by using GIS. *Environ Earth Sci* 75(10):867
- Chen W, Chai H, Sun X, Wang Q, Ding X, Hong H (2016b) A GIS-based comparative study of frequency ratio, statistical index and weights-of-evidence models in landslide susceptibility mapping. *Arabian J Geosci* 9(3):204
- Chen W, Peng J, Hong H, Shahabi H, Pradhan B, Liu J, Duan Z (2018a) Landslide susceptibility modelling using GIS-based machine learning techniques for Chongren County, Jiangxi Province, China. *Sci Total Environ* 626:1121–1135
- Chen W, Zhang S, Li R, Shahabi H (2018b) Performance evaluation of the GIS-based data mining techniques of best-first decision tree, random forest, and naïve Bayes tree for landslide susceptibility modeling. *Sci Total Environ* 644:1006–1018
- Constantin M, Bednarik M, Jurchescu MC, Vlaicu M (2011) Landslide susceptibility assessment using the bivariate statistical analysis and the index of entropy in the Sibiciu Basin (Romania). *Environ Earth Sci* 63(2):397–406
- Dikshit A, Sarkar R, Pradhan B, Segoni S, Alamri AM (2020) Rainfall induced landslide studies in Indian Himalayan Region: a critical review. *Appl Sci* 10(7):2466. <https://doi.org/10.3390/app10072466>
- Dikshit A, Pradhan B, Alamri AM (2021) Pathways and challenges of the application of artificial intelligence to geohazards modeling. *Gondwana Res* 100:290–301
- Elmoulat M, Brahim LA, Mastere M, Jemmah AI (2015) Mapping of mass movements susceptibility in the Zoumi Region using satellite image and GIS technology (Moroccan Rif). *Int J Sci Eng Res* 6(2):210–217
- Fang Z, Wang Y, Peng L, Hong H (2020) Integration of convolutional neural network and conventional machine learning classifiers for landslide susceptibility mapping. *Comput Geosci* 139:104470
- Fang Z, Wang Y, Peng L, Hong H (2021) A comparative study of heterogeneous ensemble-learning techniques for landslide susceptibility mapping. *Int J Geog Inf Sci* 35(2):321–347
- Froude MJ, Petley DN (2018) Global fatal landslide occurrence from 2004 to 2016. *Nat Hazard Earth Sys* 18(8):2161–2181
- Garosi Y, Sheklabadi M, Conoscenti C, Pourghasemi HR, Van Oost K (2019) Assessing the performance of GIS-based machine learning models with different accuracy measures for determining susceptibility to gully erosion. *Sci Total Environ* 664:1117–1132
- Gourfi A, Daoudi L, Rhoujjati A, Benkaddour A, Fagel N (2020) Use of bathymetry and clay mineralogy of reservoir sediment to reconstruct the recent changes in sediment yields from a mountain catchment in the Western High Atlas region. *Morocco Catena* 191:104560
- Grelle G, Soriano M, Revellino P, Guerriero L, Anderson MG, Dambra A, Guadagno FM (2014) Space–time prediction of rainfall-induced shallow landslides through a combined probabilistic/deterministic approach, optimized for initial water table conditions. *Bull Eng Geol Environ* 73(3):877–890
- Guo C, Montgomery DR, Zhang Y, Wang K, Yang Z (2015) Quantitative assessment of landslide susceptibility along the Xianshuihe fault zone, Tibetan Plateau, China. *Geomorphology* 248:93–110
- Hadji R, Limani Y, Baghem M, Demdoun A (2013) Geologic, topographic and climatic controls in landslide hazard assessment using GIS modeling: a case study of Souk Ahras region, NE Algeria. *Quat Int* 302:224–237
- Hollard H, Choubert G, Bronner G, Marchand J, Sougy J (1985) Carte géologique du Maroc, scale 1: 1,000,000. *Serv Carte géol Maroc*, 260(2)
- Hong H, Pourghasemi HR, Pourtaghi ZS (2016) Landslide susceptibility assessment in Lianhua County (China): a comparison between a random forest data mining technique and bivariate and multivariate statistical models. *Geomorphology* 259:105–118
- Jaafari A, Panahi M, Pham BT, Shahabi H, Bui DT, Rezaie F, Lee S (2019) Meta optimization of an adaptive neuro-fuzzy inference system with grey wolf optimizer and biogeography-based optimization algorithms for spatial prediction of landslide susceptibility. *CATENA* 175:430–445
- Jia K, Liang S, Zhang L, Wei X, Yao Y, Xie X (2014) Forest cover classification using Landsat ETM+ data and time series MODIS NDVI data. *Int J Appl Earth Obs Geoinf* 33:32–38
- Kanungo D, Arora M, Gupta R, Sarkar S (2008) Landslide risk assessment using concepts of danger pixels and fuzzy set theory in Darjeeling Himalayas. *Landslides* 5:407–416
- Kausar N, Majid A (2016) Random forest-based scheme using feature and decision levels information for multi-focus image fusion. *Pattern Anal Appl* 19(1):221–236
- Kawabata D, Bandibas J (2009) Landslide susceptibility mapping using geological data, a DEM from ASTER images and an Artificial Neural Network(ANN). *Geomorphology* 113(1–2):97–109
- Kim JC, Lee S, Jung HS, Lee S (2018) Landslide susceptibility mapping using random forest and boosted tree models in Pyeong-Chang. *Korea Geocarto Int* 33(9):1000–1015
- Kumar R, Anbalagan R (2015) Landslide susceptibility zonation in part of Tehri reservoir region using frequency ratio, fuzzy logic and GIS. *J Earth Syst Sci* 124(2):431–448
- Kumar D, Thakur M, Dubey CS, Shukla DP (2017) Landslide susceptibility mapping & prediction using support vector machine for Mandakini River Basin, Garhwal Himalaya, India. *Geomorphology* 295:115–125
- Landis JR, Koch GG (1977) The measurement of observer agreement for categorical data. *Biometrics* 159–174
- Manchar N, Benabbas C, Hadji R, Bouaicha F, Grecu F (2018) Landslide susceptibility assessment in Constantine region (NE Algeria) by means of statistical models. *Studia Geotechnica Et Mechanica* 40(3):208–219
- Mandal S, Mandal K (2018) Modeling and mapping landslide susceptibility zones using GIS based multivariate binary logistic regression (LR) model in the Rorachu river basin of eastern Sikkim Himalaya. *India Modeling Earth Syst Environ* 4:69–88
- Mathieu P (2002) Caractérisation des sols et de leurs propriétés hydrodynamiques pour la modélisation hydrologique en milieu semi-aride, Bassin versant du Tensift–Maroc, Mémoire de fin d'étude ENSAM DAA « Physique des surfaces naturelles et génie hydrologique » (ENSAR) Avril 2002–Septembre 2002
- Meinhardt M, Fink M, Tüschel H (2015) Landslide susceptibility analysis in central Vietnam based on an incomplete landslide

- inventory: comparison of a new method to calculate weighting factors by means of bivariate statistics. *Geomorphology* 234:80–97
- Merkhi A, Laftouhi NE, Soulaïmani A, Fniguire F (2015) Quantification et évaluation de l'érosion hydrique en utilisant le modèle RUSLE et dépôt intégrés dans un SIG. Application dans le bassin versant n°15 dans le haut atlas de Marrakech (Maroc). *Eur Sci J* 11(29):340–356
- Michard A, Hoepffner C, Soulaïmani A, Baidder L (2008) The variscan belt. In *Continental evolution: The geology of Morocco* (pp. 65–132). Springer, Berlin, Heidelberg
- Youssef AM, Pourghasemi HR, Pourtaghi ZS, Al-Katheeri MM (2016) Landslide susceptibility mapping using random forest, boosted regression tree, classification and regression tree, and general linear models and comparison of their performance at Wadi Tayyah Basin, Asir Region. *Saudi Arabia Landslides* 13(5):839–856
- Park S, Kim J (2019) Landslide susceptibility mapping based on random forest and boosted regression tree models, and a comparison of their performance. *Appl Sci* 9(5):942
- Park S, Hamm SY, Kim J (2019) Performance evaluation of the GIS-based data-mining techniques decision tree, random forest, and rotation forest for landslide susceptibility modeling. *Sustainability* 11(20):5659
- Pham BT, Pradhan B, Bui DT, Prakash I, Dholakia MB (2016) A comparative study of different machine learning methods for landslide susceptibility assessment: a case study of Uttarakhand area (India). *EnvironModell Software* 84:240–250
- Pham BT, Nguyen-Thoi T, Qi C, Van Phong T, Dou J, Ho LS, Prakash I (2020) Coupling RBF neural network with ensemble learning techniques for landslide susceptibility mapping. *CATENA* 195:104805
- Pourghasemi HR, Kerle N (2016) Random forests and evidential belief function-based landslide susceptibility assessment in Western Mazandaran Province. *Iran Environ Earth Sci* 75(3):185
- Zhao Y, Zhang Y (2008) Comparison of decision tree methods for finding active objects. *Adv Space Res* 41(12):1955–1959
- Pourghasemi HR, Rahmati O (2018) Prediction of the landslide susceptibility: which algorithm, which precision? *Catena* 162:177–192
- Pourghasemi HR, Moradi HR, Aghda SF (2013) Landslide susceptibility mapping by binary logistic regression, analytical hierarchy process, and statistical index models and assessment of their performances. *Nat Hazard* 69(1):749–779
- Pradhan B, Sameen MI, Al-Najjar HAH, Sheng D, Alamri AM, Park HJ (2021) A meta-learning approach of optimisation for spatial prediction of landslides. *Remote Sensing* 13(22):4521
- Rana N, Bisht P, Bagri DS, Wasson RJ, Sundriyal Y (2017) Identification of landslide-prone zones in the geomorphically and climatically sensitive Mandakini Valley, (central Himalaya), for disaster governance using the weights of evidence method. *Geomorphology* 284:41–52
- Rasyid AR, Bhandary NP, Yatabe R (2016) Performance of frequency ratio and logistic regression model in creating GIS based landslides susceptibility map at Lompobattang Mountain. *Indonesia Geoenviron Disasters* 3(1):19
- Salvatici T, Tofani V, Rossi G, D'Ambrosio M, Tacconi Stefanelli C, Casagli MEB, N, (2018) Application of a physically based model to forecast shallow landslides at a regional scale. *Nat Hazards Earth Syst Sci* 18(7):1919–1935
- Sandric I, Ionita C, Chitu Z, Dardala M, Irimia R, Furtuna FT (2019) Using CUDA to accelerate uncertainty propagation modelling for landslide susceptibility assessment. *Environ Modell Software* 115:176–186
- Santoso AM, Phoon KK, Quek ST (2011) Effects of soil spatial variability on rainfall-induced landslides. *Comput Struct* 89(11–12):893–900
- Shahabi H, Khezri S, Ahmad BB, Hashim M (2014) Landslide susceptibility mapping at central Zab basin, Iran: a comparison between analytical hierarchy process, frequency ratio and logistic regression models. *CATENA* 115:55–70
- Song Y, Niu R, Xu S, Ye R, Peng L, Guo T, Chen T (2019) Landslide susceptibility mapping based on weighted gradient boosting decision tree in Wanzhou section of the Three Gorges Reservoir Area (China). *ISPRS Int J Geo-Inf* 8(1):4
- Tsangaratos P, Ilia I (2016) Landslide susceptibility mapping using a modified decision tree classifier in the Xanthi Perfection. *Greece Landslides* 13(2):305–320
- Witten IH, Frank E (2002) Data mining: practical machine learning tools and techniques with Java implementations. *ACM SIGMOD Rec* 31(1):76–77
- Yeon YK, Han JG, Ryu KH (2010) Landslide susceptibility mapping in Injae, Korea, using a decision tree. *Eng Geol* 116(3–4):274–283
- Yilmaz I (2009) A case study from Koyulhisar (Sivas-Turkey) for landslide susceptibility mapping by artificial neural networks. *Bull Eng Geol Environ* 68(3):297–306
- Youssef AM, Pourghasemi HR (2021) Landslide susceptibility mapping using machine learning algorithms and comparison of their performance at Abha Basin, Asir Region. *Saudi Arabia Geosci Front* 12(2):639–655
- Park S, Choi C, Kim B, Kim J (2013) Landslide susceptibility mapping using frequency ratio, analytic hierarchy process, logistic regression, and artificial neural network methods at the Inje area. *Korea. Environ Earth Sci* 68(5):1443–1464
- Yusof NM, Pradhan B (2014) Landslide susceptibility mapping along PLUS expressways in Malaysia using probabilistic based model in GIS. *Iop C Ser Earth Env* 1(20):012031
- Zhang K, Wu X, Niu R, Yang K, Zhao L (2017) The assessment of landslide susceptibility mapping using random forest and decision tree methods in the Three Gorges Reservoir area. *China Environ Earth Sci* 76(11):1–20

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.